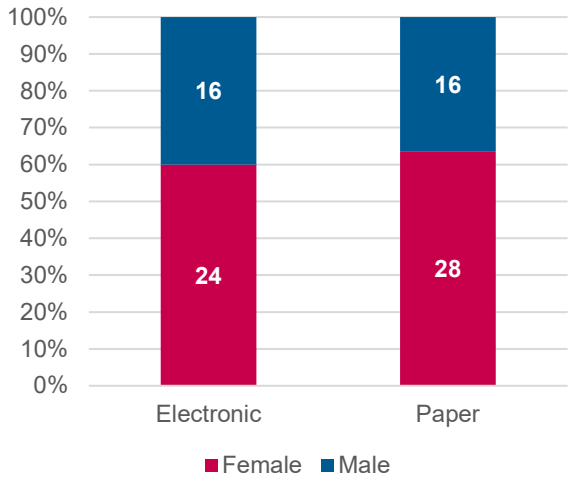
Summary Analysis

Summary Analysis

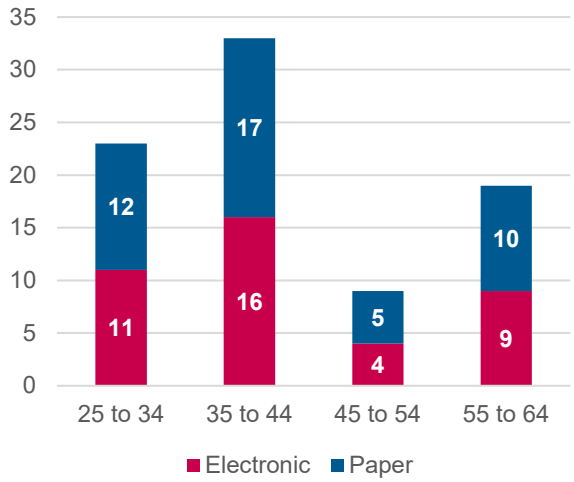
Composition of Sample Groups

Gender Differences



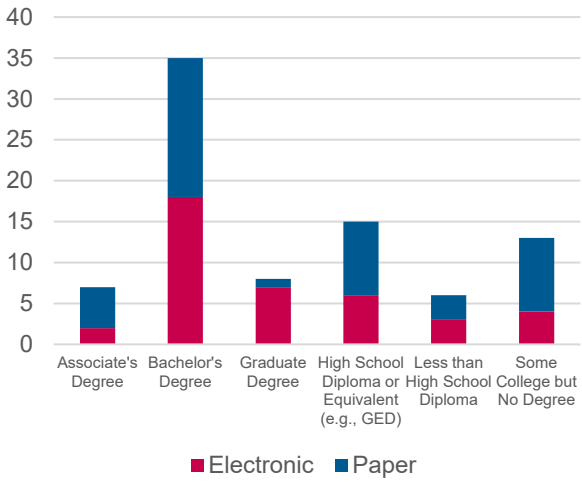
- No significant gender differences between control and treatment groups ($\chi^2 = 0.0139$, $p = 0.906$)
- Pearson's Chi Square Test with Yates' Continuity Correction used for this test

Age Differences



- No significant age differences between control and treatment groups ($\chi^2 = 0.0472$, $p = 0.997$)
- Pearson's Chi Square Test used for this test

Education Differences



- No significant age differences between control and treatment groups ($\chi^2 = 8.165$, $p = 0.142$)
- Pearson's Chi Square Test with Simulated p-Value (based on 2000 Monte Carlo simulations†) used for this test
- p-value was simulated due to the presence of expected values of less than 5

† See Bradley, D., Cutcomb, S. Monte Carlo simulations and the chi-square test of independence (1977). https://link.springer.com/content/pdf/10.3758/BF03214499.pdf

“v-Question” Analysis

& Clustering

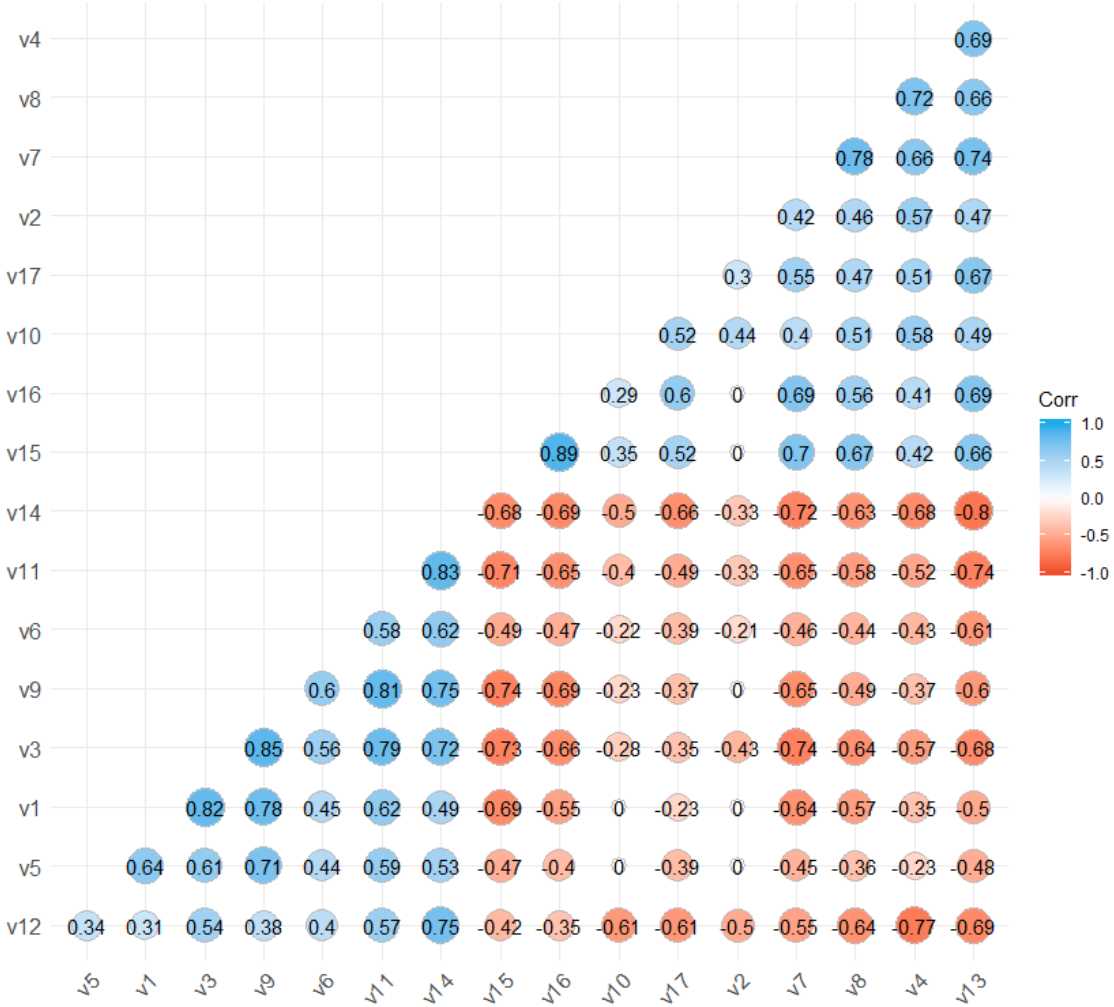
“v-Question” Analysis

Table of Questions

“v-Question” Reference Table	
Variable	**Question/Prompt**
v1	I feel pleasant
v2	I feel nervous and restless
v3	I feel satisfied with myself
v4	I wish I could be as happy as others seem to be
v5	I feel rested
v6	I am ‘calm, cool and collected'
v7	I feel that difficulties are piling up so that I cannot overcome them
v8	I worry too much over something that doesn’t really matter
v9	I am happy
v10	I have disturbing thoughts
v11	I feel secure
v12	I make decisions easily
v13	I feel inadequate
v14	I am content
v15	Unimportant thoughts run through my mind and bother me
v16	I take disappointments to heart and I can’t put them out of my mind
V17	I get in a state of tension or turmoil when I think about my recent concerns and interests

“v-Question” Analysis

Correlations Present in the Data

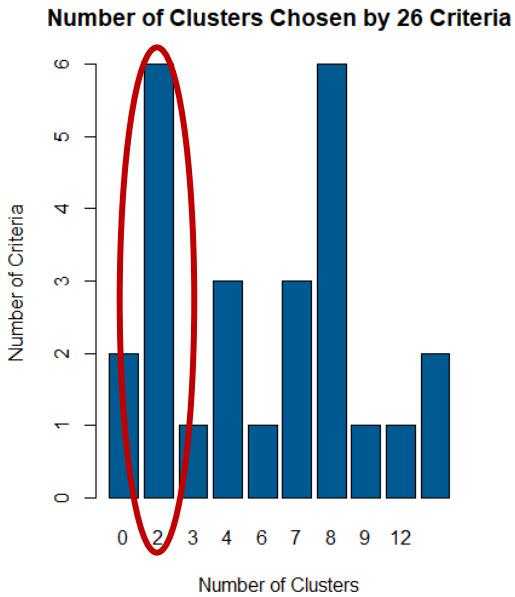


Discussion

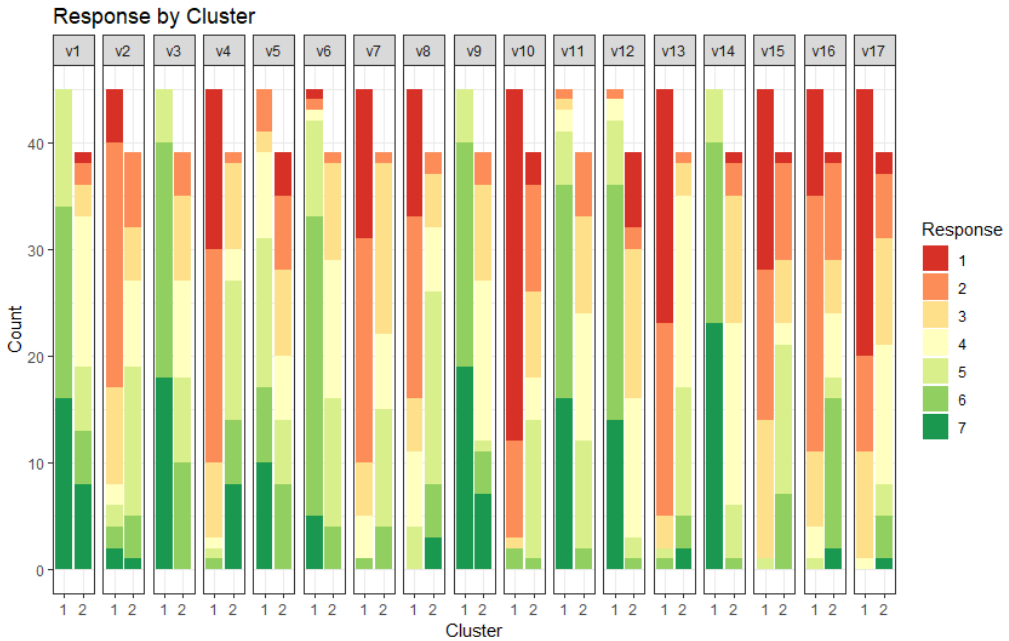
- The correlation plot to the left was constructed to aid in the visualization of correlative patterns (*Pearson’s R-Squared*) in the data.
- There appear to be consistent patterns in the data – however it is important to note that such relationships in the data do not imply real-world relationships.
- However, given the obvious nature of the relationships (the presence of unimportant, bothersome thoughts and a tendency to take disappointments to heart), it stands to reason that these differences may lend clarity to the central question by highlighting those groups whose preference for electronic information may be used effectively to reduce cost and paper waste.
- Moreover, the data may provide helpful feedback for future trials and improved consent design.

“v-Question” Analysis

Applying Unsupervised Learning to Highlight Group Differences



Charrad et al.’s NbClust package[1] was used to aid in cluster size determination. Of the 26 criteria applied, six selected 2 as the optimal number of clusters and six selected 8 as the optimal number of clusters. Due to the limited sample size, I set k=2 for 2 clusters.



The chart above illustrates responses to “v questions” by cluster assignment. There are significant differences present in responses to many of these questions (“I feel pleasant” and “I make decisions easily” for example). These differences will be explored further in an effort to uncover subtle patterns that might aid in improvements to future trial designs or present opportunities to sharpen efficacy in a targeted patient population.

As will be summarized on the following slide, there is a high degree of confidence that the two clusters do not differ in terms of age, gender, race or education composition. Thus, there appear to be no meaningful consistencies with regard to how individuals respond given their age, race, etc.

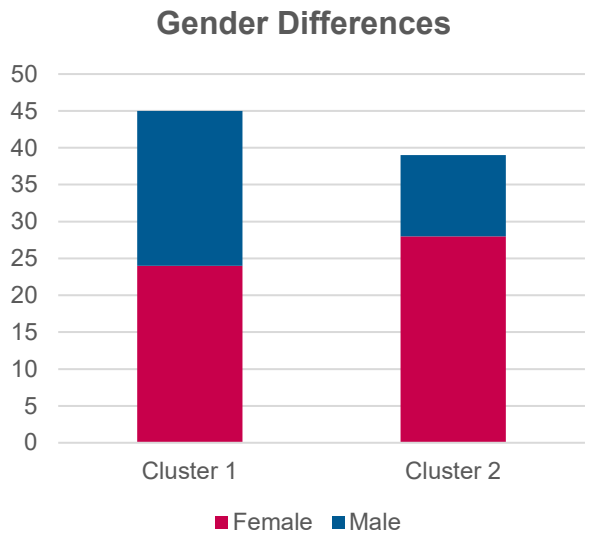
Charrad M., Ghazzali N., Boiteau V., Niknafs A. (2014). "NbClust: An R Package for Determining the Relevant Number of Clusters in a Data Set.", "Journal of Statistical Software, 61(6), 1-36.", "URL http://www.jstatsoft.org/v61/i06/".

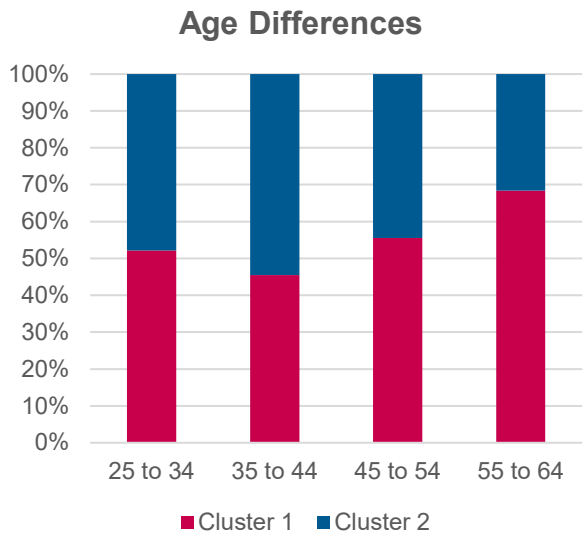
“v-Question” Analysis

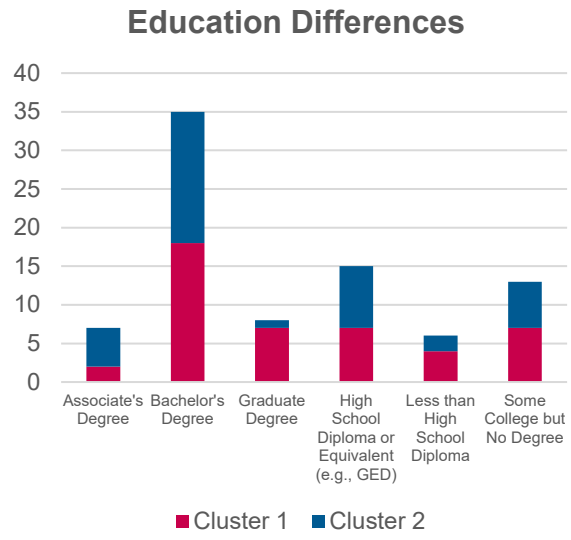
Clustered Group Differences

	Electronic	Paper
Cluster 1	20	25
Cluster 2	20	19

No significant clustering differences exist between the control and treatment groups (χ^2 = 0.1655, p = 0.684); test performed using Yates’ Continuity Correction.







- No significant gender differences between control and treatment groups (χ^2 = 2.287, p = 0.1304)
- Pearson’s Chi Square Test with Yates’ Continuity Correction used for this test

- No significant age differences between control and treatment groups (χ^2 = 2.5909, p = 0.4628)
- Pearson’s Chi Square Test with Simulated p-Value (based on 2000 Monte Carlo simulations) used for this test

- No significant age differences between control and treatment groups (χ^2 = 1.239, p = 0.919)
- Pearson’s Chi Square Test with Simulated p-Value (based on 2000 Monte Carlo simulations) used for this test

“v-Question” Analysis

Mann-Whitney U Test to Confirm Group Differences

```
Wilcoxon rank sum test with continuity correction

data:  v1[cluster == 1] and v1[cluster == 2]
W = 1326, p-value = 3.517e-05
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v2[cluster == 1] and v2[cluster == 2]
W = 401, p-value = 1.074e-05
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v3[cluster == 1] and v3[cluster == 2]
W = 1575, p-value = 8.843e-11
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v4[cluster == 1] and v4[cluster == 2]
W = 102, p-value = 1.563e-12
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v5[cluster == 1] and v5[cluster == 2]
W = 1263, p-value = 0.0004566
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v6[cluster == 1] and v6[cluster == 2]
W = 1509, p-value = 3.854e-09
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v7[cluster == 1] and v7[cluster == 2]
W = 174.5, p-value = 1.195e-10
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v8[cluster == 1] and v8[cluster == 2]
W = 197.5, p-value = 5.181e-10
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v9[cluster == 1] and v9[cluster == 2]
W = 1442, p-value = 1.728e-07
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v10[cluster == 1] and v10[cluster == 2]
W = 215.5, p-value = 4.447e-10
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v11[cluster == 1] and v11[cluster == 2]
W = 1599.5, p-value = 4.242e-11
alternative hypothesis: true location shift is not equal to 0
```

“v-Question” Analysis

Mann-Whitney U Test to Confirm Group Differences

```
Wilcoxon rank sum test with continuity correction

data:  v12[cluster == 1] and v12[cluster == 2]
W = 1682, p-value = 2.017e-13
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v13[cluster == 1] and v13[cluster == 2]
W = 80, p-value = 2.696e-13
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v14[cluster == 1] and v14[cluster == 2]
W = 1729, p-value = 6.051e-15
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v15[cluster == 1] and v15[cluster == 2]
W = 279.5, p-value = 3.959e-08
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  v17[cluster == 1] and v17[cluster == 2]
W = 224.5, p-value = 1.673e-09
alternative hypothesis: true location shift is not equal to 0
```

“s-Question” Analysis

“s-Question” Analysis

Table of Questions

*See accompanying Document for Individual Results

“s-Question” Reference Table	
Variable	**Question/Prompt**
s1	When I signed the consent form(s) for this procedure, I knew and understood that I was giving consent to participate in a procedure.
s2	The main purpose of this procedure is to collect stem cells for cancer patients through Apheresis.
s3	I was informed how long this procedure would last.
s4	As part of this research procedure, Growth Factor Injections are given.
s5	I was asked to give information about my demographics (age, education, race, gender), and understand the reason.
s6	There are no risks of harm or discomfort to me.
s7	There are risks associated with this procedure and they were clearly explained by the consent information.
s8	Risks require that I limit my travel for no less than 48 hours following the procedure.
s9	I understand that I may experience pain or discomfort following the procedure for up to one week.
s10	This a live-saving procedure that is intended to benefit cancer patients.
s11	There may not be any direct benefit to me for participating in this study.
s12	I was not offered any alternatives besides involvement in this procedure. *
s13	If I did not want to participate in this procedure, I could refuse to sign the consent form . *
s14	The anonymous results of the genetic tests performed on my family’s blood/saliva samples will be made available to the National Institutes of Health (NIH).In this study, the identity of my family is protected from use in court proceedings by a Certificate of Confidentiality.
s15	I was given information on who to contact if I have any questions about this study.
s16	The consent form that I signed described who clearly explained what I could expect during and after the procedure
s17	Once I gave consent to be in this procedure, I could not withdraw my participation.
s18	My family was not offered any alternatives besides involvement in this procedure.

“s-Question” Analysis

Question #3 Significance Report

I was informed how long this procedure would last.

Cluster	Agree	Disagree	Unsure
1	39	3	3
2	22	3	14

Pearson’s chi-squared test performed with 2000 Monte Carlo simulations reveals a potentially significant difference in the way individuals between clusters respond to this question (X2 = 11.485, p = 0.00250). To further investigate the difference, I applied the nonparametric Fisher’s Exact Test to the data (p = 0.00214).

Overall, there was no significant difference in the frequency of “Unsure” responses relative to the other questions (binomial test).

Assuming that this difference exists among the general population, one possible explanation might be that “cluster 2” individuals exhibit greater apprehension prior to the procedure and thus have greater difficulty with understanding and retaining the content.

There was no difference between control and treatment groups.

*This result may be misleading. Expected values of less than 5 may increase the likelihood of making type 1 errors.

“s-Question” Analysis

Question #4 Significance Report

As part of this research procedure, Growth Factor Injections are given.

Gender	Agree	Disagree	Unsure
Female	30	8	14
Male	18	0	14

Pearson’s chi-squared test performed with 2000 Monte Carlo simulations reveals a potentially significant difference in the way individuals between gender respond to this question ($X2 = 6.613$, $p = 0.00480$). To further investigate the difference, I applied the nonparametric Fisher’s Exact Test to the data ($p = 0.0260$).

While technically not significant ($p = 0.0533$, CI = [0.234, 0.445]), a binomial test indicates that individuals appear to respond “Unsure” frequently relative to similar questions. This could be a result of poor understanding of the purpose and role of Growth Factor among the population sampled.

There was no difference between control and treatment groups.

*This result may be misleading. Expected values of less than 5 may increase the likelihood of making type 1 errors.

“s-Question” Analysis

Question #7 Significance Report

There are risks associated with this procedure and they were clearly explained by the consent information.

Response	Agree	Disagree	Unsure
Electronic	24	8	8
Paper	33	0	11

Pearson’s chi-squared test performed with 2000 Monte Carlo simulations reveals a potentially significant difference in the way individuals between test groups respond to this question (X2 = 9.726, p = 0.00800). To further investigate the difference, I applied the nonparametric Fisher’s Exact Test to the data (p = 0.00574).

Overall, there was no significant difference in the frequency of “Unsure” responses relative to the other questions (binomial test).

Of the individuals to disagree with this statement, 75% were female – although this is unlikely to explain the effect – while there appeared to be no other meaningful patterns. It is possible that the format itself (the central question of the research) explains the potential difference. Further investigation with a larger (and possibly more geographically diverse) sample would be needed to determine the true significance of this effect.

*This result may be misleading. Expected values of less than 5 may increase the likelihood of making type 1 errors.

“s-Question” Analysis

Question #9 Significance Report

I understand that I may experience pain or discomfort following the procedure for up to one week.

Response	Agree	Disagree	Unsure
Electronic	33	4	3
Paper	34	0	10

Pearson’s chi-squared test performed with 2000 Monte Carlo simulations reveals a potentially significant difference in the way individuals between test groups respond to this question ($X2 = 7.611$, $p = 0.0160$). To further investigate the difference, I applied the nonparametric Fisher’s Exact Test to the data ($p = 0.0173$).

Overall, there was no significant difference in the frequency of “Unsure” responses relative to the other questions (binomial test).

100% of the individuals who disagreed with this statement were female, although this is unlikely to explain the effect (it was not present in the data in general).

*This result may be misleading. Expected values of less than 5 may increase the likelihood of making type 1 errors.

“s-Question” Analysis

Question #13 Significance Report

If I did not want to participate in this procedure, I could refuse to sign the consent form.

Response	Agree	Disagree	Unsure
Electronic	25	3	12
Paper	36	0	8

Pearson’s chi-squared test performed with 2000 Monte Carlo simulations reveals a potentially significant difference in the way individuals between test groups respond to this question ($X2 = 7.611$, $p = 0.0160$). To further investigate the difference, I applied the nonparametric Fisher’s Exact Test to the data ($p = 0.0173$).

Overall, there was no significant difference in the frequency of “Unsure” responses relative to the other questions (binomial test).

100% of the individuals who disagreed with this statement were female, although this is unlikely to explain the effect (it was not present in the data in general). Furthermore, the cluster each individual belonged to related significantly ($X2 = 10.563$, $p = 0.0004$) to the respond pattern.

*This result may be misleading. Expected values of less than 5 may increase the likelihood of making type 1 errors.

“s-Question” Analysis

Question #14 Significance Report

The anonymous results of the genetic tests performed on my family’s blood/saliva samples will be made available to the National Institutes of Health (NIH).In this study, the identity of my family is protected from use in court proceedings by a Certificate of Confidentiality.

Cluster	Agree	Disagree	Unsure
1	29	1	15
2	24	3	22

Pearson’s chi-squared test performed with 2000 Monte Carlo simulations reveals a potentially significant difference in the way individuals between clusters respond to this question ($X2 = 7.165$, $p = 0.0185$). To further investigate the difference, I applied the nonparametric Fisher’s Exact Test to the data ($p = 0.0201$).

Binomial test revealed a significant difference ($p < 0.001$, CI = [0.332, 0.553]), a binomial test indicates that individuals appear to respond “Unsure” frequently relative to similar questions. This could be a result of poor understanding of the purpose and role of Growth Factor among the population sampled.

There was no difference between control and treatment groups.

*This result may be misleading. Expected values of less than 5 may increase the likelihood of making type 1 errors.

“s-Question” Analysis

Question #18 Significance Report

My family was not offered any alternatives besides involvement in this procedure.

Response	Agree	Disagree	Unsure
Electronic	17	19	4
Paper	19	10	12

Pearson’s chi-squared test performed with 2000 Monte Carlo simulations reveals a potentially significant difference in the way individuals between test groups respond to this question ($X2 = 6.992$, $p = 0.0315$). To further investigate the difference, I applied the nonparametric Fisher’s Exact Test to the data ($p = 0.0349$).

Overall, there was no significant difference in the frequency of “Unsure” responses relative to the other questions (binomial test).

Nearly twice as many respondents in the treatment group disagreed with this statement as did respondents in the control group, however, three times as many individuals in the control group were unsure of how to respond. In this instance, it appears as though the electronic format may have out-performed the paper format in terms of comprehension.

“u-Question” Analysis

“u-Question” Analysis

Table of Questions

“u-Question” Reference Table	
Variable	**Question/Prompt**
u1	What types of patients benefit from procedure
u2	The process that I would undergo as part of the procedure.
u3	The fact that this procedure involves a blood collection timeframe of a few hours
u4	The fact that the platelets I gave for this study may be stored indefinitely
u5	The possible risks or discomforts that I might experience in the procedure
u6	The possible benefits of participating in the procedure
u7	How participation in this study may benefit other children
u8	What the researchers are trying to learn from the study
u9	The effect of participation on my confidential medical information
u10	Who would pay for and provide medical treatment if I became injured or ill from participation in the procedure
u11	Whom I should contact if I have questions or concerns about the procedure
u12	The fact that participation in this procedure is voluntary
u13	Overall, how well did you understand this study?

“u-Question” Analysis

Mann-Whitney U Test to Confirm Group Differences

```
Wilcoxon rank sum test with continuity correction

data:  u1[cluster == 1] and u1[cluster == 2]
W = 1025, p-value = 0.04707
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  u2[cluster == 1] and u2[cluster == 2]
W = 1125.5, p-value = 0.003557
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  u3[cluster == 1] and u3[cluster == 2]
W = 1025, p-value = 0.03305
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  u4[cluster == 1] and u4[cluster == 2]
W = 1131.5, p-value = 0.002888
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  u5[cluster == 1] and u5[cluster == 2]
W = 1112, p-value = 0.007062
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  u6[cluster == 1] and u6[cluster == 2]
W = 1115, p-value = 0.007136
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  u7[cluster == 1] and u7[cluster == 2]
W = 1143, p-value = 0.01139
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  u8[cluster == 1] and u8[cluster == 2]
W = 1312, p-value = 5.921e-05
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  u9[cluster == 1] and u9[cluster == 2]
W = 1135, p-value = 0.01315
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  u10[cluster == 1] and u10[cluster == 2]
W = 1129, p-value = 0.02077
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  u11[cluster == 1] and u11[cluster == 2]
W = 1289, p-value = 0.0001211
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  u12[cluster == 1] and u12[cluster == 2]
W = 1073, p-value = 0.04204
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  u13[cluster == 1] and u13[cluster == 2]
W = 1220.5, p-value = 0.001349
alternative hypothesis: true location shift is not equal to 0
```

“u-Question” Analysis

Question #8 Significance Report

What the researchers are trying to learn from the study.

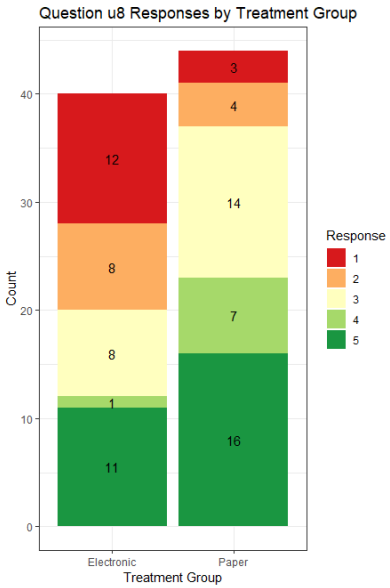
When a Wilcoxon rank sum test with continuity correction (also known as Mann-Whitney U) was applied to this data, there was a significant difference (W = 1169.5, p = 0.0076) in the median response values between the control group (median of 4) and the treatment group (median of 2.5).

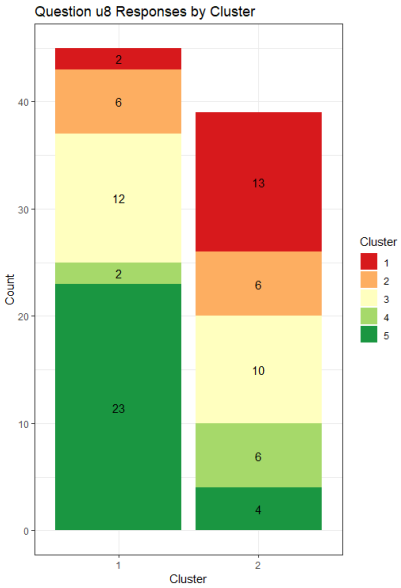
Response	1	2	3	4	5
Electronic	12	8	8	1	11
Paper	3	4	14	7	16

Distribution of responses by treatment group

Cluster	1	2	3	4	5
1	2	6	12	2	23
2	13	6	10	6	4

Distribution of responses by cluster





“u-Question” Analysis

Question #10 Significance Report

Who would pay for and provide medical treatment if I became injured or ill from participation in the procedure.

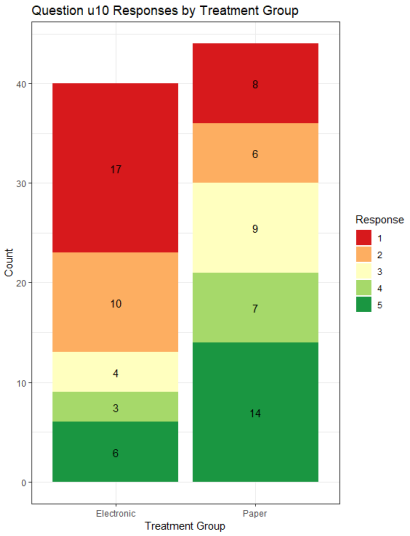
When a Wilcoxon rank sum test with continuity correction (also known as Mann-Whitney U) was applied to this data, there was a significant difference (W = 1206.5, p = 0.0027) in the median response values between the control group (median of 3) and the treatment group (median of 2).

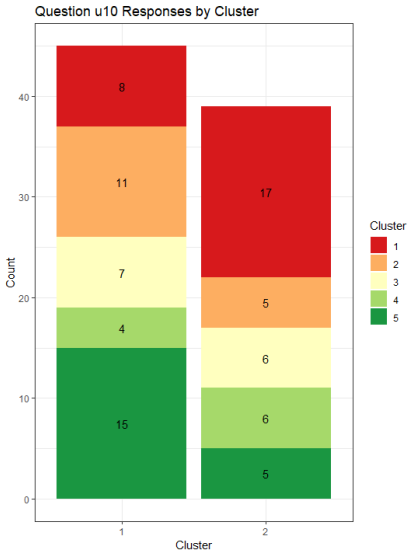
Response	1	2	3	4	5
Electronic	17	10	4	3	6
Paper	8	6	9	7	14

Distribution of responses by treatment group

Cluster	1	2	3	4	5
1	1	8	11	4	15
2	17	5	6	6	5

Distribution of responses by cluster





“u-Question” Analysis

Mann-Whitney U Test Results for Control vs. Treatment (α = 0.05)*

Results of Mann-Whitney U Test			Results of t-Test		
Question	W	p-Value	Question	t	p-Value
1	748	0.329	1	-0.844	0.401
2	912	0.467	2	1.356	0.179
3	853.5	0.882	3	0.466	0.642
4	860	0.843	4	0.655	0.514
5	914	0.469	5	1.168	0.247
6	924.5	0.413	6	1.175	0.244
7	976.5	0.360	7	1.232	0.222
8	1169.5	0.00758	8	2.805	0.00642
9	939	0.573	9	1.187	0.239
10	1206.5	0.00271	10	3.146	0.00231
11	1086.5	0.0544	11	2.0163	0.0472
12	950.5	0.466	12	1.662	0.101
13	1018	0.198	13	1.407	0.164

*Mann-Whitney U and t-tests were performed to minimize both type I and type II errors, although the distribution of the data did not meet the necessary criteria for the t-test to yield reliable results (Shapiro-Wilks normality test applied to all 13 questions; results differed significantly from normality for all 13).

“n-Question” Analysis

“n-Question” Analysis

Table of Questions

“n-Question” Reference Table	
Variable	**Question/Prompt**
n1	1. I know which options are available to me.
n2	2. I know the benefits of each option.
n3	3. I know the risks and side effects of each option.
n4	4. I am clear about which benefits matter most to me.
n5	5. I am clear about which risks and side effects matter most.
n6	6. I am clear about which is more important to me (the benefits or the risks and side effects).
n7	7. I have enough support from others to make a choice.
n8	8. I am choosing without pressure from others.
n9	9. I have enough advice to make a choice.
n10	10. I am clear about the best choice for me.
n11	11. I feel sure about what to choose.
n12	12. This decision is easy for me to make.
n13	13. I feel I have made an informed choice.
n14	14. My decision shows what is important to me.
n15	15. I expect to stick with my decision.
n16	16. I am satisfied with my decision.

“n-Question” Analysis

Mann-Whitney U Test to Confirm Group Differences

```
Wilcoxon rank sum test with continuity correction

data:  n1[cluster == 1] and n1[cluster == 2]
W = 1263.5, p-value = 6.826e-05
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n2[cluster == 1] and n2[cluster == 2]
W = 1209, p-value = 0.0007063
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n3[cluster == 1] and n3[cluster == 2]
W = 1275.5, p-value = 6.697e-05
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n4[cluster == 1] and n4[cluster == 2]
W = 1248, p-value = 0.0001673
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n5[cluster == 1] and n5[cluster == 2]
W = 1238, p-value = 0.0002967
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n6[cluster == 1] and n6[cluster == 2]
W = 1213, p-value = 0.0006332
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n7[cluster == 1] and n7[cluster == 2]
W = 1289, p-value = 2.622e-05
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n8[cluster == 1] and n8[cluster == 2]
W = 1271, p-value = 5.78e-05
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n9[cluster == 1] and n9[cluster == 2]
W = 1292.5, p-value = 6.43e-05
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n10[cluster == 1] and n10[cluster == 2]
W = 1230, p-value = 0.0008269
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n11[cluster == 1] and n11[cluster == 2]
W = 1253, p-value = 0.0003786
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n12[cluster == 1] and n12[cluster == 2]
W = 1176.5, p-value = 0.004757
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n13[cluster == 1] and n13[cluster == 2]
W = 1302.5, p-value = 8.19e-05
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n14[cluster == 1] and n14[cluster == 2]
W = 1106, p-value = 0.0166
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n15[cluster == 1] and n15[cluster == 2]
W = 1175, p-value = 0.001225
alternative hypothesis: true location shift is not equal to 0
```

```
Wilcoxon rank sum test with continuity correction

data:  n16[cluster == 1] and n16[cluster == 2]
W = 1249.5, p-value = 0.000251
alternative hypothesis: true location shift is not equal to 0
```

“n-Question” Analysis

Mann-Whitney U Test Results for Control vs. Treatment (No Significant Results, α = 0.05)*

Results of Mann-Whitney U Test			Results of t-Test		
Question	W	p-Value	Question	t	p-Value
1	686	0.926	1	0.083	0.934
2	927	0.512	2	0.812	0.419
3	970	0.291	3	1.204	0.232
4	888	0.778	4	0.579	0.564
5	987.5	0.223	5	1.389	0.169
6	893	0.742	6	0.595	0.554
7	679.5	0.085	7	-2.010	0.048
8	747	0.176	8	-1.476	0.144
9	846.5	0.751	9	-0.302	0.763
10	925.5	0.669	10	0.674	0.503
11	853	0.802	11	-0.082	0.935
12	682	0.062	12	-1.789	0.077
13	964	0.439	13	0.896	0.373
14	815	0.685	14	0.152	0.879
15	965.5	0.356	15	0.776	0.440
16	878.5	0.992	16	0.230	0.819

*Mann-Whitney U and t-tests were performed to minimize both type I and type II errors, although the distribution of the data did not meet the necessary criteria for the t-test to yield reliable results (Shapiro-Wilks normality test applied to all 16 questions; results differed significantly from normality for all 16). Given the nature of question 7, I decided not to perform further analysis.